Discriminant Analysis

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1. Bayes Theorem

Let - $d \in \mathbb{N}^*$ be number of features

- $K \in \mathbb{N}^*$ such that $K \geq 2$, be number of classes
- $\mathcal{S} = \{1, \dots, K\}$
- Y a categorical random variable

1.1. Discrete case

Let $(X_1, \dots, X_d)$ be a discrete random vector

Given $x = (x_1, \dots, x_d) \in \text{supp}(X) \subset \mathbb{R}^d$ and $y \in \mathcal{S}$.

The bayes theorem gives that

$$\mathbb{P}(Y = y | X_1 = x_1, \dots, X_d = x_d) = \frac{\mathbb{P}(Y = y)\mathbb{P}(X_1 = x_1, \dots, X_d = x_d | Y = y)}{\mathbb{P}(X_1 = x_1, \dots, X_d = x_d)}$$

i.e.

$$\mathbb{P}(y | x_1, \dots, x_d) = \frac{\mathbb{P}(y)\mathbb{P}(x_1, \dots, x_d | y)}{\mathbb{P}(x_1, \dots, x_d)}$$

(if we alleviate the notations)

1.2. Continuous case

Let $X = (X_1, \dots, X_d)$ be a real random vector

- $\pi_y = \mathbb{P}(Y = y)$ be the prior probability of class y
- $f_{X|Y}(x|y)$ be the conditional density of X given $Y = y$
- $f_X(x) = \sum_{k \in \mathcal{S}} \pi_k f_{X|Y}(x|k)$ be the marginal density of X

Given $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ and $y \in \mathcal{S}$

The Bayes theorem gives that

$$\mathbb{P}(Y = y | X = x) = \frac{\pi_y f_{X|Y}(x|y)}{f_X(x)} = \frac{\pi_y f_{X|Y}(x|y)}{\sum_{k \in \mathcal{S}} \pi_k f_{X|Y}(x|k)}$$

i.e.

$$\mathbb{P}(y | x_1, \dots, x_d) = \frac{\pi_y f(x_1, \dots, x_d | y)}{f(x_1, \dots, x_d)}$$

(if we alleviate the notations)

2. Models definition

2.1. Quadratic Discriminant Analysis

Let $X = (X_1, \dots, X_d) \in \mathbb{R}^d$ be a random vector

Given $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ and $y \in \mathcal{S}$.

We suppose that:

$$X|Y = y \in \mathbb{R}^d \sim \mathcal{N}(\mu_y, \Sigma_y) \quad \text{such that} \quad \mu_y \in \mathbb{R}^d \quad \text{and} \quad \Sigma_y \in \mathbb{R}^{d\times d}, \Sigma_y \succ 0 \text{ and is symmetric}$$

i.e. that the random variable $X|Y = y$ has a multivariate Gaussian distribution with density

$$\forall x \in \mathbb{R}^d, \quad f_{X|Y}(x|y) = \frac{1}{(2\pi)^{\frac{d}{2}}|\Sigma_y|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu_y)^T\Sigma_y^{-1}(x-\mu_y)\right)$$

2.2. Linear Discriminant Analysis

If we suppose

$$\forall y \in \mathcal{S}, \Sigma_y = \Sigma$$

i.e. the covariance of features is the same in each class, this model is called **Linear Discriminant Analysis**

3. Gaussian density parameters estimation

Let

- $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$ be a training set of n i.i.d. observations, with $x_i \in \mathbb{R}^d$ and $y_i \in \mathcal{S}$
- $\mathcal{I}_y = \{i \in \{1, \dots, n\} | y_i = y\}$ be the set of indices of the observations of class y
- $n_y = |\mathcal{I}_y|$ be the number of observations of class y

The parameters to estimate are the prior probabilities π_y, the means μ_y and the covariance matrices Σ_y.

For the matrix formulations below, let

- $X \in \mathbb{R}^{n\times d}$ be the matrix whose i^{th} row is x_i
- $X_y \in \mathbb{R}^{n_y\times d}$ be the matrix whose rows are the observations of class y (ordered as in $\mathcal{I}_y$)
- $\mathbf{1}_{n_y} \in \mathbb{R}^{n_y}$ be the all-ones vector

3.1. Prior probabilities

The priors are estimated by the empirical frequencies of the classes:

$$\hat{\pi}_y = \frac{n_y}{n} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{y_i = y\}}$$

3.2. Means

The mean of each class is estimated by the empirical mean of the observations of that class:

$$\hat{\mu}_y = \frac{1}{n_y}\sum_{i\in\mathcal{I}_y} x_i = \frac{1}{n_y}\mathbf{1}_{n_y}^T X_y$$

3.3. Covariance matrices (QDA)

The covariance matrix of each class is estimated by the empirical covariance of the observations of that class:

$$\hat{\Sigma}_y = \frac{1}{n_y} \sum_{i \in \mathcal{I}_y} (x_i - \hat{\mu}_y)(x_i - \hat{\mu}_y)^T$$

In matrix form:

$$\hat{\Sigma}_y = \frac{1}{n_y} (X_y - \mathbf{1}_{n_y} \hat{\mu}_y^T)^T (X_y - \mathbf{1}_{n_y} \hat{\mu}_y^T)$$

3.4. Weighted average Covariance matrix (LDA)

Under LDA, all classes share the same covariance matrix. It is estimated by computing a weighted average of covariances from each class:

$$\hat{\Sigma} = \frac{1}{n} \sum_{y \in \mathcal{S}} \sum_{i \in \mathcal{I}_y} (x_i - \hat{\mu}_y)(x_i - \hat{\mu}_y)^T = \frac{1}{n} \sum_{y \in \mathcal{S}} n_y \hat{\Sigma}_y$$

In matrix form, with the within-class scatter matrix S_w:

$$\hat{\Sigma} = \frac{1}{n} S_w = \frac{1}{n} \sum_{y \in \mathcal{S}} (X_y - \mathbf{1}_{n_y} \hat{\mu}_y^T)^T (X_y - \mathbf{1}_{n_y} \hat{\mu}_y^T)$$

The estimates $\hat{\mu}_y$ and $\hat{\Sigma}_y$ above are the maximum likelihood estimates. Unbiased versions are obtained by replacing n_y with $n_y - 1$ (and n with $n - K$ for the pooled covariance), although the difference vanishes as $n_y \to \infty$.

4. Discriminant functions

4.1. Classifier

Given $x = (x_1, \dots, x_d) \in \mathbb{R}^d$.

The Bayes optimal classifier is such that:

$$g^*(x) = \underset{y \in \mathcal{S}}{\operatorname{argmax}}\ f(y | x_1, \dots, x_d) = \underset{y \in \mathcal{S}}{\operatorname{argmax}}\ \frac{\mathbb{P}(y) f(x_1, \dots, x_d | y)}{f(x_1, \dots, x_d)}$$

Since $f(x_1, \dots, x_d)$ does not depend on y, the Bayes classifier is such that

$$g^*(x) = \underset{y \in \mathcal{S}}{\operatorname{argmax}}\ \mathbb{P}(y) f(x_1, \dots, x_d | y)$$

4.2. Discriminant functions

Starting from the Bayes classifier of §4.1

$$g^*(x) = \underset{y \in \mathcal{S}}{\operatorname{argmax}}\ \pi_y f_{X|Y}(x|y)$$

with $\pi_y = \mathbb{P}(Y = y)$. Since log is strictly increasing, applying it does not change the argmax:

$$\log \left(\pi_y f_{X|Y}(x|y) \right) = \log \pi_y + \log f_{X|Y}(x|y)$$

Using the Gaussian density of §2.1:

$$\log f_{X|Y}(x|y) = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_y| - \frac{1}{2}(x-\mu_y)^T\Sigma_y^{-1}(x-\mu_y)$$

The term $-\frac{d}{2}\log(2\pi)$ does not depend on y and can be dropped, which defines the **discriminant function**

$$\delta_y(x) = \log\pi_y - \frac{1}{2}\log|\Sigma_y| - \frac{1}{2}(x-\mu_y)^T\Sigma_y^{-1}(x-\mu_y)$$

such that

$$g^*(x) = \underset{y\in\mathcal{S}}{\operatorname{argmax}}\ \delta_y(x)$$

4.2.1. Quadratic case

Expanding the quadratic form:

$$(x-\mu_y)^T\Sigma_y^{-1}(x-\mu_y) = x^T\Sigma_y^{-1}x - 2\mu_y^T\Sigma_y^{-1}x + \mu_y^T\Sigma_y^{-1}\mu_y$$

so that

$$\delta_y(x) = \log\pi_y - \frac{1}{2}\log|\Sigma_y| - \frac{1}{2}\left(x^T\Sigma_y^{-1}x - 2\mu_y^T\Sigma_y^{-1}x + \mu_y^T\Sigma_y^{-1}\mu_y\right)$$

which is a quadratic function in x. The decision boundary between two classes $k, l \in \mathcal{S}$ is given by $\delta_k(x) = \delta_l(x)$, a quadratic surface.

4.2.2. Linear case

Under LDA, $\Sigma_y = \Sigma$ for all $y \in \mathcal{S}$. The terms $-\frac{1}{2}x^T\Sigma^{-1}x$ and $-\frac{1}{2}\log|\Sigma|$ then do not depend on y.

So

$$\underset{y\in\mathcal{S}}{\operatorname{argmax}}\ \delta_y(x) = \underset{y\in\mathcal{S}}{\operatorname{argmax}}\ \tilde{\delta}_y(x)$$

With

$$\tilde{\delta}_y(x) = \log\pi_y + \frac{1}{2}\left(2\mu_y^T\Sigma^{-1}x - \mu_y^T\Sigma^{-1}\mu_y\right)$$

which is linear in x. The decision boundary between two classes $k, l \in \mathcal{S}$ is the hyperplane

$$\delta_k(x) = \delta_l(x) \iff (\mu_k - \mu_l)^T\Sigma^{-1}x = \frac{1}{2}\left(\mu_k^T\Sigma^{-1}\mu_k - \mu_l^T\Sigma^{-1}\mu_l\right) - \log\frac{\pi_k}{\pi_l}$$

5. Examples

Actual vs Predicted Classes on a couple datasets

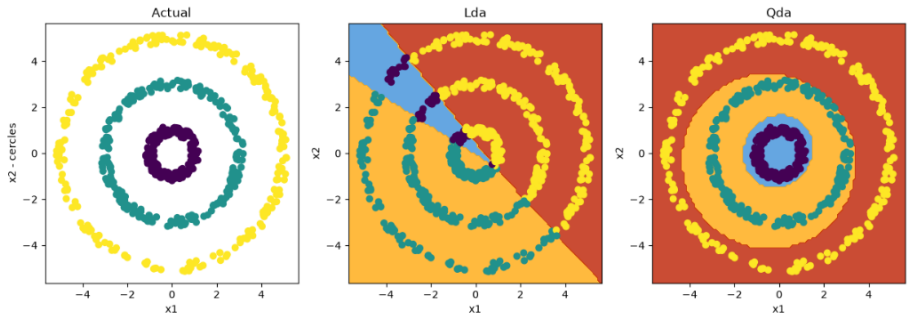


Figure 1: Cercles

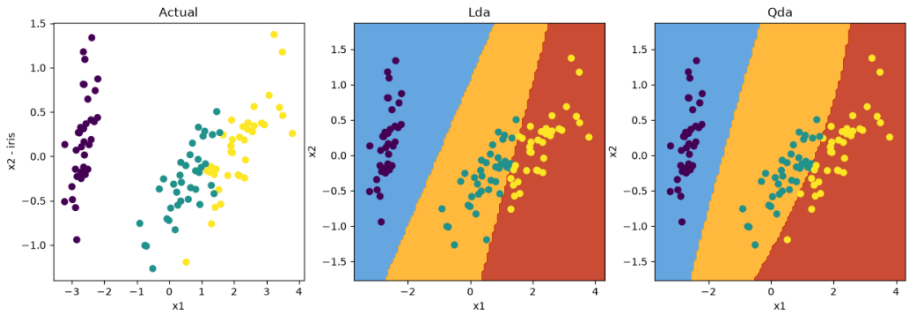


Figure 2: Iris

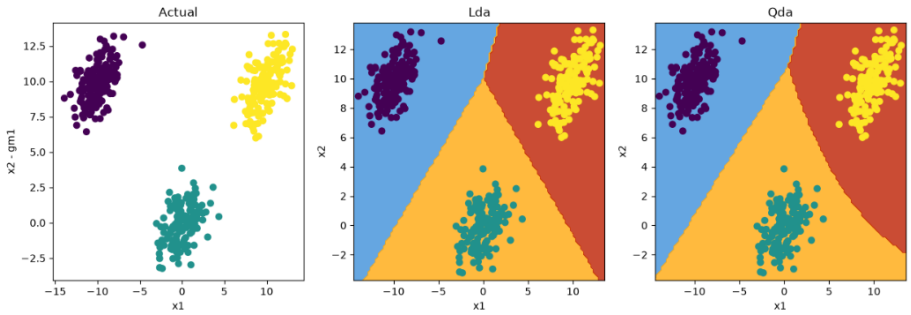


Figure 3: Gaussian mixture (shared covariance)

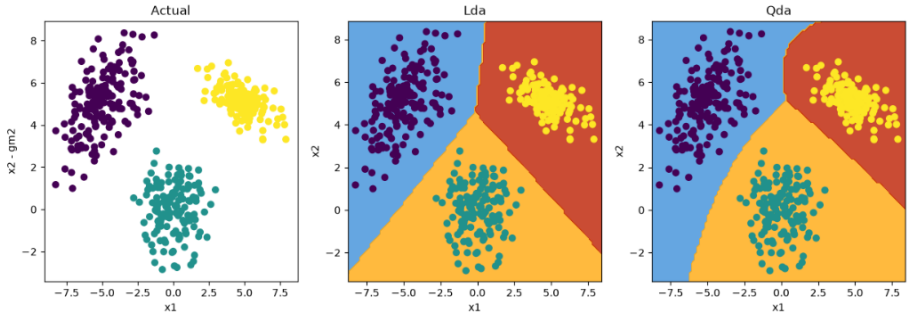


Figure 4: Gaussian mixture (class-specific covariance)